

Laboratorium 10.2 — GLMM (NBA, skuteczność rzutów wolnych)

Statystyka w SI i analizie danych — materiał i zadanie z 8-glmm-2.html

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2026-05-19

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Pakiety: dplyr, ggplot2, lme4 (glmer); dane przez utils::read.csv (bez readr).

Plik danych: szukany jest nba-players-2022-3.csv w kolejności: params\$nba_csv (jeśli ustawisz w YAML / rmarkdown::render(..., params = list(nba_csv = \"ścieżka\")), potem 110/2/, 110/, 110/data/. Gdy **nigdzie go nie ma**, dokument wczyta **zbiór demonstracyjny** (ten sam kod dalej działa; na zajęciach podmień na prawdziwy CSV z wykładu).

```
library(dplyr)
library(ggplot2)
library(lme4)
```

Przykład — osiągnięcia graczy NBA

```
candidate_paths <- c(
  if (!is.null(params$nba_csv) && nzchar(as.character(params$nba_csv))) {
    as.character(params$nba_csv)
  } else {
    character()
  },
  "nba-players-2022-3.csv",
  "../nba-players-2022-3.csv",
  "../data/nba-players-2022-3.csv"
)
nba_csv <- candidate_paths[file.exists(candidate_paths)][1]

if (is.na(nba_csv)) {
  message(
    "Brak pliku nba-players-2022-3.csv - używam syntetycznych danych demonstracyjnych ",
    "(do raportu z zajęć skopiuj CSV z materiałów do 110/2/ lub 110/)."
```

```

n <- 180L
rk <- seq_len(n)
pos_lab <- rep(c("PG", "SG", "SF-SG", "PF", "C"), length.out = n)
pos1 <- sub("^([[:alnum:]]+).*", "\\1", pos_lab)
g <- sample(20:82, n, replace = TRUE)
pts <- pmax(1L, rpois(n, lambda = 8L + as.integer(as.factor(pos1)) * 2L))
FTA <- pmax(2L, rpois(n, lambda = 5L))
eta <- -1.2 + 0.04 * (pts / g) + 0.15 * (as.integer(as.factor(pos1)) - 3)
p_fta <- pmin(0.92, pmax(0.45, plogis(eta)))
FT <- rbinom(n, FTA, p_fta)
nba_df <- data.frame(
  Rk = rk,
  Player = sprintf("Player_%04d", rk),
  Pos = pos_lab,
  Tm = sample(c("BOS", "LAL", "CHI", "MIA", "DAL"), n, replace = TRUE),
  FT = FT,
  FTA = FTA,
  PTS = pts,
  G = g,
  stringsAsFactors = FALSE
)
nba_df[["FT%"]] <- ifelse(nba_df$FTA > 0, 100 * nba_df$FT / nba_df$FTA, NA_real_)
} else {
  message("Wczytuje: ", normalizePath(nba_csv, winslash = "/", mustWork = FALSE))
  nba_df <- read.csv(
    nba_csv,
    stringsAsFactors = FALSE,
    check.names = FALSE,
    fileEncoding = "UTF-8"
  )
}
head(nba_df)

```

```

##      Rk      Player   Pos  Tm FT  FTA PTS  G      FT%
## 1    1 Player_0001    PG CHI  0   2  15 20   0.00000
## 2    2 Player_0002    SG LAL  6   6  25 42 100.00000
## 3    3 Player_0003 SF-SG MIA  5   6  12 46  83.33333
## 4    4 Player_0004    PF MIA  2   6  15 47  33.33333
## 5    5 Player_0005     C LAL  2   5  11 77  40.00000
## 6    6 Player_0006    PG BOS  2   5  12 56  40.00000

```

```
nba_df <- nba_df |> filter(if_all(everything(), ~ !is.na(.)))
```

```

nba_df |>
  select(Rk, Player, Pos, Tm) |>
  filter(grepl("-", Pos, fixed = TRUE))

```

```

##      Rk      Player   Pos  Tm
## 1     3 Player_0003 SF-SG MIA
## 2     8 Player_0008 SF-SG CHI
## 3    13 Player_0013 SF-SG BOS
## 4    18 Player_0018 SF-SG MIA
## 5    23 Player_0023 SF-SG MIA
## 6    28 Player_0028 SF-SG MIA
## 7    33 Player_0033 SF-SG BOS

```

```
## 8 38 Player_0038 SF-SG CHI
## 9 43 Player_0043 SF-SG CHI
## 10 48 Player_0048 SF-SG DAL
## 11 53 Player_0053 SF-SG DAL
## 12 58 Player_0058 SF-SG CHI
## 13 63 Player_0063 SF-SG BOS
## 14 68 Player_0068 SF-SG CHI
## 15 73 Player_0073 SF-SG DAL
## 16 78 Player_0078 SF-SG MIA
## 17 83 Player_0083 SF-SG MIA
## 18 88 Player_0088 SF-SG DAL
## 19 93 Player_0093 SF-SG LAL
## 20 98 Player_0098 SF-SG CHI
## 21 103 Player_0103 SF-SG CHI
## 22 108 Player_0108 SF-SG BOS
## 23 113 Player_0113 SF-SG LAL
## 24 118 Player_0118 SF-SG BOS
## 25 123 Player_0123 SF-SG CHI
## 26 128 Player_0128 SF-SG BOS
## 27 133 Player_0133 SF-SG BOS
## 28 138 Player_0138 SF-SG CHI
## 29 143 Player_0143 SF-SG MIA
## 30 148 Player_0148 SF-SG LAL
## 31 153 Player_0153 SF-SG LAL
## 32 158 Player_0158 SF-SG DAL
## 33 163 Player_0163 SF-SG BOS
## 34 168 Player_0168 SF-SG CHI
## 35 173 Player_0173 SF-SG DAL
## 36 178 Player_0178 SF-SG LAL
```

```
nba_df <-
  nba_df |>
  mutate(Pos = as.factor(sub("^([[:alnum:]]+).*", "\\1", Pos)))
table(nba_df$Pos)
```

```
##
## C PF PG SF SG
## 36 36 36 36 36
```

```
transferred_df <-
  nba_df |>
  select(Rk, Tm) |>
  group_by(Rk) |>
  summarise(NClubs = n(), .groups = "drop") |>
  filter(NClubs > 1)
```

```
transferred_df
```

```
## # A tibble: 0 x 2
## # i 2 variables: Rk <int>, NClubs <int>
```

```
nba_df <-
  nba_df |>
  filter(!(Rk %in% transferred_df$Rk) | Tm == "TOT")
nba_df
```

##	Rk	Player	Pos	Tm	FT	FTA	PTS	G	FT%
## 1	1	Player_0001	PG	CHI	0	2	15	20	0.00000
## 2	2	Player_0002	SG	LAL	6	6	25	42	100.00000
## 3	3	Player_0003	SF	MIA	5	6	12	46	83.33333
## 4	4	Player_0004	PF	MIA	2	6	15	47	33.33333
## 5	5	Player_0005	C	LAL	2	5	11	77	40.00000
## 6	6	Player_0006	PG	BOS	2	5	12	56	40.00000
## 7	7	Player_0007	SG	DAL	1	2	20	65	50.00000
## 8	8	Player_0008	SF	CHI	1	3	16	66	33.33333
## 9	9	Player_0009	PF	LAL	2	5	17	82	40.00000
## 10	10	Player_0010	C	CHI	2	7	16	47	28.57143
## 11	11	Player_0011	PG	CHI	4	13	11	24	30.76923
## 12	12	Player_0012	SG	LAL	3	5	19	61	60.00000
## 13	13	Player_0013	SF	BOS	4	8	15	24	50.00000
## 14	14	Player_0014	PF	MIA	1	5	5	54	20.00000
## 15	15	Player_0015	C	MIA	1	5	11	66	20.00000
## 16	16	Player_0016	PG	DAL	0	4	16	27	0.00000
## 17	17	Player_0017	SG	LAL	2	7	18	53	28.57143
## 18	18	Player_0018	SF	MIA	2	4	17	58	50.00000
## 19	19	Player_0019	PF	LAL	0	3	11	55	0.00000
## 20	20	Player_0020	C	MIA	2	5	12	44	40.00000
## 21	21	Player_0021	PG	CHI	1	3	20	74	33.33333
## 22	22	Player_0022	SG	MIA	4	5	15	53	80.00000
## 23	23	Player_0023	SF	MIA	2	5	15	52	40.00000
## 24	24	Player_0024	PF	MIA	0	2	5	82	0.00000
## 25	25	Player_0025	C	LAL	2	2	11	42	100.00000
## 26	26	Player_0026	PG	CHI	1	6	10	39	16.66667
## 27	27	Player_0027	SG	DAL	3	4	13	61	75.00000
## 28	28	Player_0028	SF	MIA	4	7	11	53	57.14286
## 29	29	Player_0029	PF	DAL	2	3	15	49	66.66667
## 30	30	Player_0030	C	DAL	0	2	7	73	0.00000
## 31	31	Player_0031	PG	LAL	4	6	10	44	66.66667
## 32	32	Player_0032	SG	BOS	2	4	17	22	50.00000
## 33	33	Player_0033	SF	BOS	6	6	16	73	100.00000
## 34	34	Player_0034	PF	BOS	2	5	12	53	40.00000
## 35	35	Player_0035	C	LAL	3	7	10	54	42.85714
## 36	36	Player_0036	PG	CHI	2	5	18	77	40.00000
## 37	37	Player_0037	SG	MIA	0	3	20	53	0.00000
## 38	38	Player_0038	SF	CHI	1	2	18	54	50.00000
## 39	39	Player_0039	PF	MIA	2	2	5	57	100.00000
## 40	40	Player_0040	C	LAL	3	7	9	67	42.85714
## 41	41	Player_0041	PG	LAL	2	6	3	77	33.33333
## 42	42	Player_0042	SG	LAL	2	3	15	44	66.66667
## 43	43	Player_0043	SF	CHI	2	4	13	44	50.00000
## 44	44	Player_0044	PF	MIA	2	8	14	44	25.00000
## 45	45	Player_0045	C	BOS	1	5	13	51	20.00000
## 46	46	Player_0046	PG	BOS	3	4	13	66	75.00000
## 47	47	Player_0047	SG	MIA	2	6	22	75	33.33333
## 48	48	Player_0048	SF	DAL	2	4	26	22	50.00000
## 49	49	Player_0049	PF	LAL	1	3	16	25	33.33333
## 50	50	Player_0050	C	BOS	5	7	11	31	71.42857
## 51	51	Player_0051	PG	DAL	1	6	16	61	16.66667
## 52	52	Player_0052	SG	LAL	1	4	21	35	25.00000
## 53	53	Player_0053	SF	DAL	0	2	16	59	0.00000

## 54	54	Player_0054	PF	LAL	2	2	12	24	100.00000
## 55	55	Player_0055	C	LAL	3	7	14	27	42.85714
## 56	56	Player_0056	PG	MIA	5	7	15	82	71.42857
## 57	57	Player_0057	SG	MIA	3	5	22	72	60.00000
## 58	58	Player_0058	SF	CHI	1	8	17	46	12.50000
## 59	59	Player_0059	PF	CHI	1	4	14	42	25.00000
## 60	60	Player_0060	C	MIA	7	7	5	43	100.00000
## 61	61	Player_0061	PG	DAL	4	4	13	49	100.00000
## 62	62	Player_0062	SG	LAL	0	4	13	40	0.00000
## 63	63	Player_0063	SF	BOS	2	2	12	64	100.00000
## 64	64	Player_0064	PF	CHI	7	8	16	48	87.50000
## 65	65	Player_0065	C	DAL	0	2	14	39	0.00000
## 66	66	Player_0066	PG	LAL	2	6	12	77	33.33333
## 67	67	Player_0067	SG	LAL	4	8	21	34	50.00000
## 68	68	Player_0068	SF	CHI	1	5	23	55	20.00000
## 69	69	Player_0069	PF	DAL	1	3	5	48	33.33333
## 70	70	Player_0070	C	LAL	1	4	12	54	25.00000
## 71	71	Player_0071	PG	LAL	2	6	11	53	33.33333
## 72	72	Player_0072	SG	DAL	2	5	20	75	40.00000
## 73	73	Player_0073	SF	DAL	2	9	16	49	22.22222
## 74	74	Player_0074	PF	DAL	3	8	14	57	37.50000
## 75	75	Player_0075	C	MIA	5	9	12	37	55.55556
## 76	76	Player_0076	PG	BOS	3	3	9	77	100.00000
## 77	77	Player_0077	SG	DAL	4	7	15	21	57.14286
## 78	78	Player_0078	SF	MIA	1	2	12	38	50.00000
## 79	79	Player_0079	PF	CHI	4	6	15	38	66.66667
## 80	80	Player_0080	C	CHI	3	8	12	22	37.50000
## 81	81	Player_0081	PG	MIA	3	6	14	59	50.00000
## 82	82	Player_0082	SG	LAL	3	6	19	33	50.00000
## 83	83	Player_0083	SF	MIA	0	2	14	23	0.00000
## 84	84	Player_0084	PF	MIA	0	4	10	48	0.00000
## 85	85	Player_0085	C	LAL	3	8	9	57	37.50000
## 86	86	Player_0086	PG	DAL	1	4	15	70	25.00000
## 87	87	Player_0087	SG	BOS	2	4	19	35	50.00000
## 88	88	Player_0088	SF	DAL	4	5	20	78	80.00000
## 89	89	Player_0089	PF	DAL	1	4	14	61	25.00000
## 90	90	Player_0090	C	BOS	2	2	11	64	100.00000
## 91	91	Player_0091	PG	LAL	0	2	15	48	0.00000
## 92	92	Player_0092	SG	DAL	4	6	18	60	66.66667
## 93	93	Player_0093	SF	LAL	1	3	19	43	33.33333
## 94	94	Player_0094	PF	LAL	3	8	13	68	37.50000
## 95	95	Player_0095	C	LAL	1	4	13	43	25.00000
## 96	96	Player_0096	PG	BOS	1	2	10	51	50.00000
## 97	97	Player_0097	SG	DAL	0	2	14	25	0.00000
## 98	98	Player_0098	SF	CHI	3	4	20	56	75.00000
## 99	99	Player_0099	PF	CHI	1	4	12	25	25.00000
## 100	100	Player_0100	C	BOS	2	2	11	76	100.00000
## 101	101	Player_0101	PG	CHI	2	4	4	78	50.00000
## 102	102	Player_0102	SG	LAL	0	3	15	72	0.00000
## 103	103	Player_0103	SF	CHI	1	7	13	52	14.28571
## 104	104	Player_0104	PF	BOS	2	3	10	35	66.66667
## 105	105	Player_0105	C	LAL	3	6	17	47	50.00000
## 106	106	Player_0106	PG	BOS	3	4	17	58	75.00000
## 107	107	Player_0107	SG	CHI	3	4	25	29	75.00000

##	108	108	Player_0108	SF	BOS	1	2	11	78	50.00000
##	109	109	Player_0109	PF	BOS	1	3	10	34	33.33333
##	110	110	Player_0110	C	CHI	0	2	8	67	0.00000
##	111	111	Player_0111	PG	CHI	4	6	14	79	66.66667
##	112	112	Player_0112	SG	CHI	3	4	20	61	75.00000
##	113	113	Player_0113	SF	LAL	4	8	20	24	50.00000
##	114	114	Player_0114	PF	MIA	2	4	16	32	50.00000
##	115	115	Player_0115	C	MIA	1	3	11	64	33.33333
##	116	116	Player_0116	PG	CHI	2	5	14	38	40.00000
##	117	117	Player_0117	SG	BOS	3	7	34	65	42.85714
##	118	118	Player_0118	SF	BOS	5	7	13	63	71.42857
##	119	119	Player_0119	PF	LAL	2	7	11	76	28.57143
##	120	120	Player_0120	C	BOS	0	4	9	36	0.00000
##	121	121	Player_0121	PG	BOS	5	5	19	20	100.00000
##	122	122	Player_0122	SG	LAL	1	5	25	51	20.00000
##	123	123	Player_0123	SF	CHI	2	4	13	76	50.00000
##	124	124	Player_0124	PF	DAL	2	5	13	25	40.00000
##	125	125	Player_0125	C	BOS	4	13	12	79	30.76923
##	126	126	Player_0126	PG	LAL	1	2	9	71	50.00000
##	127	127	Player_0127	SG	LAL	1	3	12	21	33.33333
##	128	128	Player_0128	SF	BOS	1	3	17	69	33.33333
##	129	129	Player_0129	PF	DAL	1	6	14	21	16.66667
##	130	130	Player_0130	C	BOS	3	6	4	77	50.00000
##	131	131	Player_0131	PG	BOS	2	7	17	21	28.57143
##	132	132	Player_0132	SG	LAL	2	2	17	70	100.00000
##	133	133	Player_0133	SF	BOS	1	4	14	22	25.00000
##	134	134	Player_0134	PF	DAL	4	8	11	32	50.00000
##	135	135	Player_0135	C	LAL	1	4	16	67	25.00000
##	136	136	Player_0136	PG	BOS	0	2	10	63	0.00000
##	137	137	Player_0137	SG	LAL	3	10	24	22	30.00000
##	138	138	Player_0138	SF	CHI	2	3	11	32	66.66667
##	139	139	Player_0139	PF	MIA	3	6	9	28	50.00000
##	140	140	Player_0140	C	CHI	1	2	5	70	50.00000
##	141	141	Player_0141	PG	DAL	1	3	14	42	33.33333
##	142	142	Player_0142	SG	CHI	4	7	15	44	57.14286
##	143	143	Player_0143	SF	MIA	2	4	15	60	50.00000
##	144	144	Player_0144	PF	CHI	3	7	14	23	42.85714
##	145	145	Player_0145	C	BOS	2	4	8	40	50.00000
##	146	146	Player_0146	PG	MIA	3	6	13	50	50.00000
##	147	147	Player_0147	SG	MIA	1	3	19	75	33.33333
##	148	148	Player_0148	SF	LAL	2	5	23	24	40.00000
##	149	149	Player_0149	PF	MIA	4	7	11	26	57.14286
##	150	150	Player_0150	C	MIA	0	3	5	53	0.00000
##	151	151	Player_0151	PG	BOS	2	4	14	65	50.00000
##	152	152	Player_0152	SG	MIA	5	10	14	67	50.00000
##	153	153	Player_0153	SF	LAL	0	5	11	77	0.00000
##	154	154	Player_0154	PF	DAL	4	6	11	75	66.66667
##	155	155	Player_0155	C	BOS	5	5	10	25	100.00000
##	156	156	Player_0156	PG	DAL	4	5	10	40	80.00000
##	157	157	Player_0157	SG	CHI	0	3	19	66	0.00000
##	158	158	Player_0158	SF	DAL	3	6	9	51	50.00000
##	159	159	Player_0159	PF	LAL	3	4	18	76	75.00000
##	160	160	Player_0160	C	CHI	3	3	5	22	100.00000
##	161	161	Player_0161	PG	LAL	2	4	11	68	50.00000

```
## 162 162 Player_0162 SG DAL 2 6 15 24 33.33333
## 163 163 Player_0163 SF BOS 4 4 13 45 100.00000
## 164 164 Player_0164 PF DAL 1 5 11 65 20.00000
## 165 165 Player_0165 C DAL 4 7 13 71 57.14286
## 166 166 Player_0166 PG MIA 4 8 15 45 50.00000
## 167 167 Player_0167 SG CHI 0 3 18 69 0.00000
## 168 168 Player_0168 SF CHI 3 4 11 40 75.00000
## 169 169 Player_0169 PF DAL 2 6 12 52 33.33333
## 170 170 Player_0170 C CHI 2 3 8 48 66.66667
## 171 171 Player_0171 PG CHI 2 5 13 60 40.00000
## 172 172 Player_0172 SG CHI 2 5 16 46 40.00000
## 173 173 Player_0173 SF DAL 2 3 15 41 66.66667
## 174 174 Player_0174 PF CHI 2 5 9 58 40.00000
## 175 175 Player_0175 C BOS 2 2 10 21 100.00000
## 176 176 Player_0176 PG CHI 1 3 14 43 33.33333
## 177 177 Player_0177 SG CHI 2 7 15 35 28.57143
## 178 178 Player_0178 SF LAL 2 4 20 55 50.00000
## 179 179 Player_0179 PF LAL 5 7 12 75 71.42857
## 180 180 Player_0180 C LAL 2 3 11 27 66.66667
```

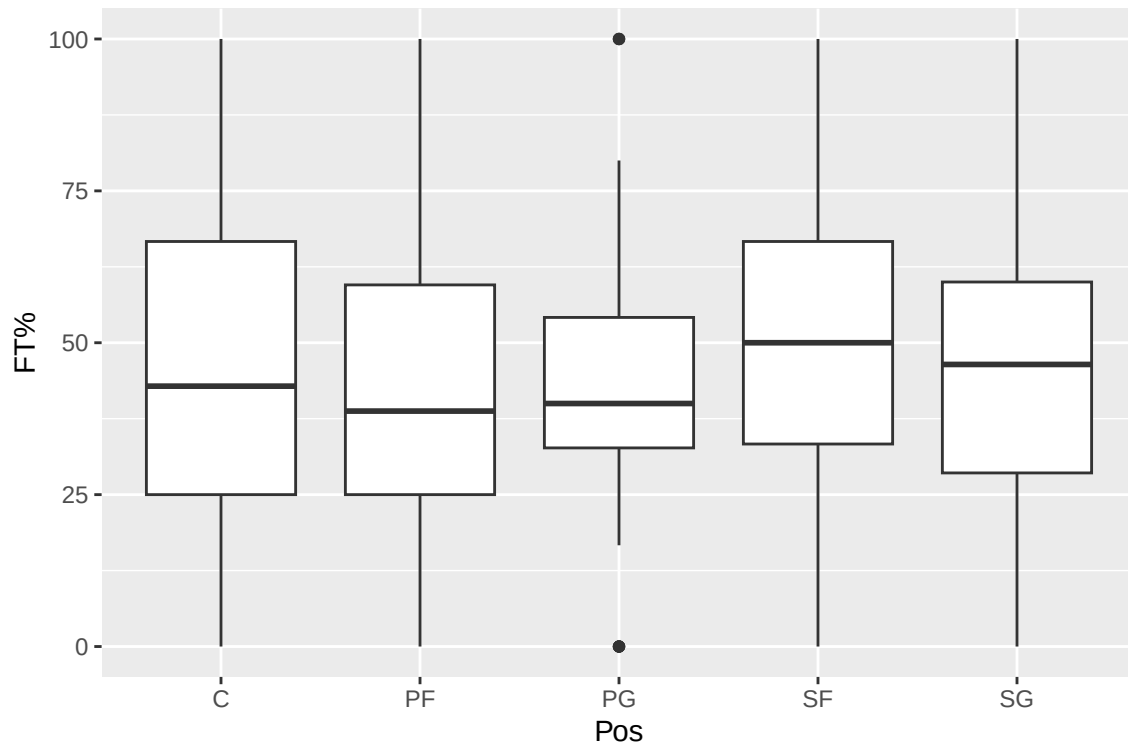
```
nba_df |>
  group_by(Pos) |>
  summarise(Acc = sum(FT) / sum(FTA), .groups = "drop")
```

```
## # A tibble: 5 x 2
##   Pos      Acc
##   <fct> <dbl>
## 1 C      0.463
## 2 PF     0.429
## 3 PG     0.457
## 4 SF     0.482
## 5 SG     0.449
```

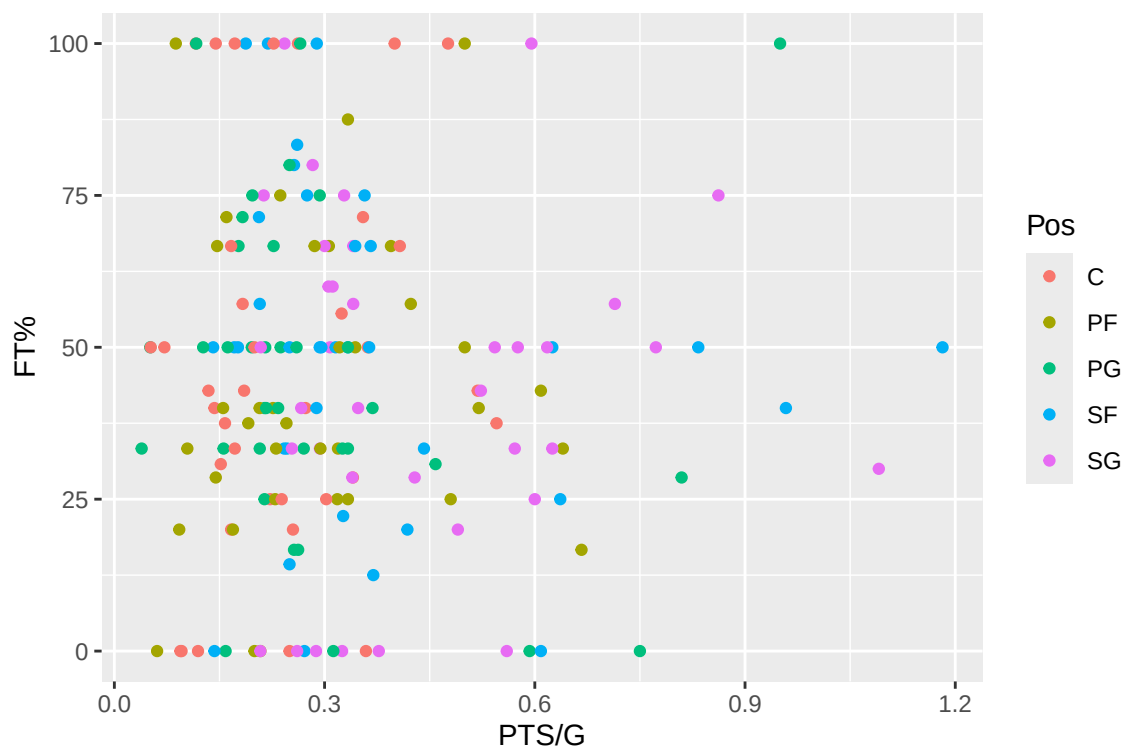
```
nba_df |>
  summarise(TotalAcc = sum(FT) / sum(FTA))
```

```
##   TotalAcc
## 1 0.4552752
```

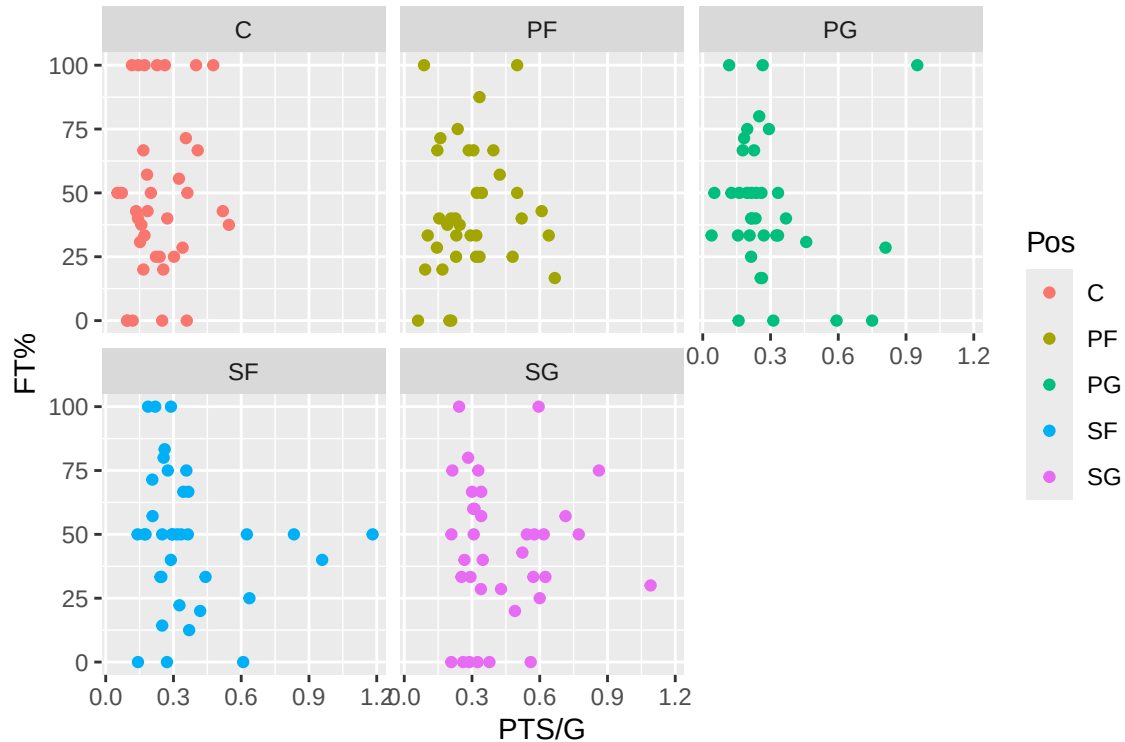
```
nba_df |>
  ggplot(aes(x = Pos, y = `FT%`)) +
  geom_boxplot()
```



```
nba_df |>
  ggplot(aes(x = PTS / G, y = `FT%`)) +
  geom_point(aes(color = Pos))
```



```
nba_df |>
  ggplot(aes(x = PTS / G, y = `FT%`)) +
  geom_point(aes(color = Pos)) +
  facet_wrap(~Pos)
```



Analiza GLMM — model RI (materiał)

Model z losowym wyrazem wolnym dla pozycji oraz dla zagnieżdżenia gracz–pozycja (logit, odpowiedź dwumianowa: trafione i chybite rzuty wolne).

```
nba_fit <- glmer(
  cbind(FT, FTA - FT) ~ 1 +
    (1 | Pos) +
    (1 | Pos:Player),
  data = nba_df,
  family = binomial(link = "logit")
)
summary(nba_fit)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(FT, FTA - FT) ~ 1 + (1 | Pos) + (1 | Pos:Player)
## Data: nba_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    553.7    563.3   -273.9    547.7      177
##
## Scaled residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -1.8017 -0.7335 -0.1123  0.6733  2.3793
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## Pos:Player (Intercept) 0.1264   0.3555
## Pos      (Intercept) 0.0000   0.0000
## Number of obs: 180, groups: Pos:Player, 180; Pos, 5
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.18327    0.07389   -2.48   0.0131 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
fe <- fixef(nba_fit)
fes <- c(fe, plogis(fe))
names(fes)[2] <- "FT_Acc"
fes
```

```
## (Intercept)      FT_Acc
## -0.1832732    0.4543095
```

```
predict(nba_fit, re.form = NA, type = "response")
```

```
##      1      2      3      4      5      6      7      8
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      9     10     11     12     13     14     15     16
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     17     18     19     20     21     22     23     24
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     25     26     27     28     29     30     31     32
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     33     34     35     36     37     38     39     40
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     41     42     43     44     45     46     47     48
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     49     50     51     52     53     54     55     56
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     57     58     59     60     61     62     63     64
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     65     66     67     68     69     70     71     72
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     73     74     75     76     77     78     79     80
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     81     82     83     84     85     86     87     88
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     89     90     91     92     93     94     95     96
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##     97     98     99    100    101    102    103    104
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##    105    106    107    108    109    110    111    112
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##    113    114    115    116    117    118    119    120
```

```
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      121      122      123      124      125      126      127      128
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      129      130      131      132      133      134      135      136
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      137      138      139      140      141      142      143      144
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      145      146      147      148      149      150      151      152
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      153      154      155      156      157      158      159      160
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      161      162      163      164      165      166      167      168
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      169      170      171      172      173      174      175      176
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
##      177      178      179      180
## 0.4543095 0.4543095 0.4543095 0.4543095
```

```
ranef(nba_fit)$Pos |>
  mutate(FT_Acc = plogis(`(Intercept)` + fe[[1]]))
```

```
##      (Intercept)      FT_Acc
## C              0 0.4543095
## PF              0 0.4543095
## PG              0 0.4543095
## SF              0 0.4543095
## SG              0 0.4543095
```

```
predict(
  nba_fit,
  nba_df |> distinct(Pos) |> arrange(Pos),
  re.form = ~(1 | Pos),
  type = "response"
)
```

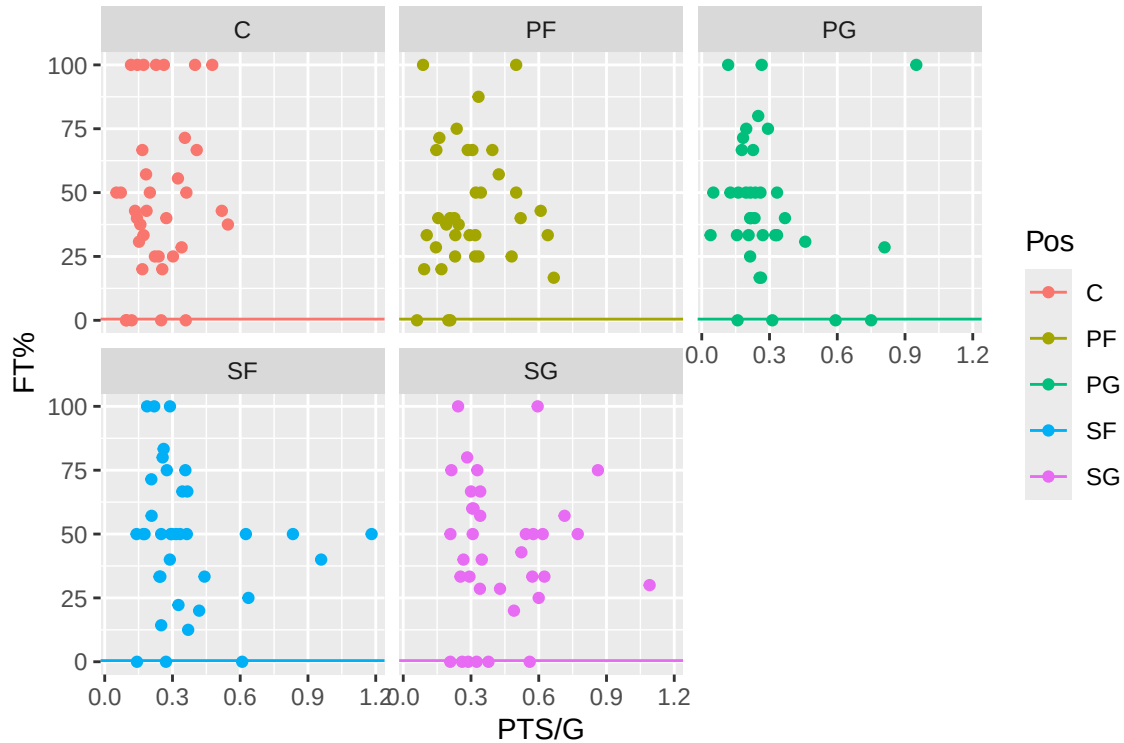
```
##           1           2           3           4           5
## 0.4543095 0.4543095 0.4543095 0.4543095 0.4543095
```

```
nba_df |>
  group_by(Pos) |>
  summarise(FT_Acc_MLE = sum(FT) / sum(FTA), .groups = "drop") |>
  mutate(
    FT_Acc_GLMM = predict(nba_fit, data.frame(Pos = Pos), re.form = ~(1 | Pos), type = "response")
  )
```

```
## # A tibble: 5 x 3
##   Pos   FT_Acc_MLE FT_Acc_GLMM
##   <fct>       <dbl>       <dbl>
## 1 C         0.463         0.454
## 2 PF        0.429         0.454
## 3 PG        0.457         0.454
## 4 SF        0.482         0.454
## 5 SG        0.449         0.454
```

```
nba_df |>
  ggplot(aes(x = PTS / G, y = `FT%`)) +
  geom_point(aes(color = Pos)) +
```

```
geom_hline(
  aes(
    yintercept = predict(nba_fit, data.frame(Pos = Pos), re.form = ~(1 | Pos), type = "response"),
    color = Pos
  )
) +
facet_wrap(~Pos)
```



```
player_rks <- c(2, 143, 488, 101, 137, 348, 164, 397, 58, 335, 24, 149)
nba_df |>
  filter(Rk %in% player_rks) |>
  mutate(
    Predicted = predict(nba_fit, data.frame(Player = Player, Pos = Pos), type = "response"),
    PosPredicted = predict(nba_fit, data.frame(Pos = Pos), re.form = ~(1 | Pos), type = "response")
  ) |>
  select(Rk, Player, Pos, Predicted, `FT%`, FT, FTA, PosPredicted) |>
  arrange(Pos)
```

##	Rk	Player	Pos	Predicted	FT%	FT	FTA	PosPredicted
## 1	24	Player_0024	PF	0.4276693	0.00000	0	2	0.4543095
## 2	149	Player_0149	PF	0.4754318	57.14286	4	7	0.4543095
## 3	164	Player_0164	PF	0.4201029	20.00000	1	5	0.4543095
## 4	101	Player_0101	PG	0.4594019	50.00000	2	4	0.4543095
## 5	58	Player_0058	SF	0.3892568	12.50000	1	8	0.4543095
## 6	143	Player_0143	SF	0.4594019	50.00000	2	4	0.4543095
## 7	2	Player_0002	SG	0.5410826	100.00000	6	6	0.4543095
## 8	137	Player_0137	SG	0.4177382	30.00000	3	10	0.4543095

Zadanie — porównanie modeli GLMM

Treść (z materiału): porównaj model RI (`nba_fit`) z:

1. uproszczonym modelem RI z efektem losowym tylko na poziomie pozycji (`1 | Pos`);
2. modelem z drugim efektem stałym — średnią liczbą punktów na mecz (`PTS/G`) — po **ustandaryzowaniu**, oraz z losowymi wyrazami wolnymi jak w modelu wyjściowym;
3. modelem z dwoma efektami stałymi jak w pkt. 2 oraz **trzecią** strukturą losową: losowe nachylenie ustandaryzowanej średniej punktów na mecz na poziomie pozycji, tj. łącznie dwa efekty stałe (`intercept + scale_pts`) i trzy składowe losowe: (`1 | Pos`), (`1 | Pos:Player`) oraz (`0 + scale_pts | Pos`) (interpretacja: pozycje różnią się nie tylko poziomem, ale i związkiem skuteczności z typową „produktywnością” punktową).

```
nba_df <- nba_df |>
  mutate(
    pts_pg = PTS / G,
    scale_pts = as.numeric(scale(pts_pg))
  )
```

```
nba_ri_pos_only <- glmer(
  cbind(FT, FTA - FT) ~ 1 + (1 | Pos),
  data = nba_df,
  family = binomial(link = "logit")
)
summary(nba_ri_pos_only)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(FT, FTA - FT) ~ 1 + (1 | Pos)
## Data: nba_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  553.5    559.9   -274.8    549.5      178
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.0442 -0.8244 -0.1419  0.7352  2.8940
##
## Random effects:
## Groups Name          Variance Std.Dev.
## Pos      (Intercept)  0          0
## Number of obs: 180, groups: Pos, 5
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.1794    0.0680  -2.638  0.00834 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## optimizer (Nelder-Mead) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
nba_ri_pts <- glmer(
  cbind(FT, FTA - FT) ~ scale_pts +
    (1 | Pos) +
    (1 | Pos:Player),
```

```

data = nba_df,
family = binomial(link = "logit")
)
summary(nba_ri_pts)

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(FT, FTA - FT) ~ scale_pts + (1 | Pos) + (1 | Pos:Player)
## Data: nba_df
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##       555.7       568.4     -273.8     547.7      176
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8196 -0.7328 -0.1107  0.6619  2.3603
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## Pos:Player (Intercept) 1.252e-01 3.538e-01
## Pos          (Intercept) 2.034e-10 1.426e-05
## Number of obs: 180, groups:  Pos:Player, 180; Pos, 5
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.18217    0.07393  -2.464   0.0137 *
## scale_pts   -0.02089    0.07163  -0.292   0.7706
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale_pts -0.048
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

ctrl <- glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 5e5))
nba_ri_pts_pos_slope <- glmer(
  cbind(FT, FTA - FT) ~ scale_pts +
    (1 | Pos / Player) +
    (0 + scale_pts | Pos),
  data = nba_df,
  family = binomial(link = "logit"),
  control = ctrl
)
summary(nba_ri_pts_pos_slope)

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(FT, FTA - FT) ~ scale_pts + (1 | Pos/Player) + (0 + scale_pts |
## Pos)
## Data: nba_df

```

```
## Control: ctrl
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    557.7    573.6   -273.8    547.7      175
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8196 -0.7328 -0.1107  0.6619  2.3602
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## Player.Pos (Intercept) 0.1252   0.3538
## Pos        (Intercept) 0.0000   0.0000
## Pos.1      scale_pts   0.0000   0.0000
## Number of obs: 180, groups: Player:Pos, 180; Pos, 5
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.18217    0.07393  -2.464   0.0137 *
## scale_pts   -0.02089    0.07163  -0.292   0.7706
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale_pts -0.048
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
aic_tab <- data.frame(
  model = c(
    "RI (materiał): (1|Pos) + (1|Pos:Player)",
    "RI uproszczony: tylko (1|Pos)",
    "RI + scale(PTS/G) stały",
    "RI + scale(PTS/G) + losowe nachylenie scale_pts | Pos"
  ),
  AIC = c(
    AIC(nba_fit),
    AIC(nba_ri_pos_only),
    AIC(nba_ri_pts),
    AIC(nba_ri_pts_pos_slope)
  ),
  BIC = c(
    BIC(nba_fit),
    BIC(nba_ri_pos_only),
    BIC(nba_ri_pts),
    BIC(nba_ri_pts_pos_slope)
  )
)
```

```
knitr::kable(aic_tab, digits = 2, caption = "Porównanie kryteriami informacyjnymi (niżej lepiej, przy p
```

Table 1: Porównanie kryteriami informacyjnymi (niżej lepiej, przy porównywalnym celu predykcyjnym).

model	AIC	BIC
RI (materiał): (1 Pos) + (1 Pos:Player)	553.74	563.32
RI uproszczony: tylko (1 Pos)	553.54	559.92
RI + scale(PTS/G) stały	555.66	568.43
RI + scale(PTS/G) + losowe nachylenie scale_pts Pos	557.66	573.62

Testy LR dla zagnieżdżonych par (dokładność asymptotyczna):

```
anova(nba_ri_pos_only, nba_fit, test = "Chisq")
```

```
## Data: nba_df
## Models:
## nba_ri_pos_only: cbind(FT, FTA - FT) ~ 1 + (1 | Pos)
## nba_fit: cbind(FT, FTA - FT) ~ 1 + (1 | Pos) + (1 | Pos:Player)
##               npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## nba_ri_pos_only    2 553.54 559.92 -274.77    549.54
## nba_fit            3 553.74 563.32 -273.87    547.74 1.7985  1    0.1799
```

```
anova(nba_fit, nba_ri_pts, test = "Chisq")
```

```
## Data: nba_df
## Models:
## nba_fit: cbind(FT, FTA - FT) ~ 1 + (1 | Pos) + (1 | Pos:Player)
## nba_ri_pts: cbind(FT, FTA - FT) ~ scale_pts + (1 | Pos) + (1 | Pos:Player)
##               npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## nba_fit            3 553.74 563.32 -273.87    547.74
## nba_ri_pts         4 555.66 568.43 -273.83    547.66 0.0832  1    0.7731
```

Krótką interpretacją (do uzupełnienia własnym tekstem na zajęcia): czy uproszczenie struktury losowej (tylko pozycja) znacząco psuje dopasowanie względem pełnego RI? Czy dodanie ustandaryzowanej średniej punktów na mecz poprawia model? Czy trzeci składnik losowy (nachylenia na poziomie Pos) jest potrzebny w świetle AIC/BIC i ewentualnego `anova(...)` dla sensownej pary zagnieżdżonej?